Problem 4

A 4-digit PIN is formed by choosing 4 digits, where each digit can be any of $0, 1, \ldots, 9$. Let $X =$ the number of nonzero digits in the PIN.

The possible values of X are determined by how many of the 4 digits are nonzero.

$$X \in \{0, 1, 2, 3, 4\}.$$

This is because:

- $X = 0$: all 4 digits are 0,
- $X = 1$: exactly 1 digit is nonzero,
- $X = 2$: exactly 2 digits are nonzero,
- $X = 3$: exactly 3 digits are nonzero,
- $X = 4$: all 4 digits are nonzero.

Three possible outcomes and their corresponding values of X are:

$$0000 \quad \Rightarrow \quad X = 0$$

$$1050 \quad \Rightarrow \quad X = 2$$

$$4827 \quad \Rightarrow \quad X = 4$$

Answer: The possible values of X are

$$\boxed{0, 1, 2, 3, 4}.$$

Three example outcomes are 0000 with $X = 0$, 1050 with $X = 2$, and 4827 with $X = 4$.

Problem 6

Each car is observed until the first car that turns left (L) appears. Let $X =$ the number of cars observed.

Since the experiment stops at the first L, an outcome must consist of any number of A's and/or R's, followed by one L.

Thus the possible values of X are

$$X \in \{1, 2, 3, \ldots\}.$$

That is,

- $X = 1$ if the first car turns left,
- $X = 2$ if the first car is not L and the second car is L,
- $X = 3$ if the first two cars are not L and the third car is L,
- and so on.

Five possible outcomes and their associated values of X are:

$$L \quad \Rightarrow \quad X = 1$$

$$AL \quad \Rightarrow \quad X = 2$$

$$RL \quad \Rightarrow \quad X = 2$$

$$AAL \quad \Rightarrow \quad X = 3$$

$$RRAL \quad \Rightarrow \quad X = 4$$

Answer: The possible values of X are

$$\boxed{1, 2, 3, \ldots}$$

and five example outcomes are L, AL, RL, AAL, and $RRAL$, with associated values $1, 2, 2, 3$, and 4, respectively.

Problem 10

Let

$$S_1 = \text{number of pumps in use at the 6-pump station,}$$
$$S_2 = \text{number of pumps in use at the 4-pump station.}$$

Then

$$S_1 \in \{0, 1, 2, 3, 4, 5, 6\}, \qquad S_2 \in \{0, 1, 2, 3, 4\}.$$

(a) $T =$ the total number of pumps in use.

$$T = S_1 + S_2$$

The smallest possible total is $0 + 0 = 0$, and the largest is $6 + 4 = 10$. Hence

$$T \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Answer:

$$\boxed{\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}}$$

(b) $X =$ the difference between the numbers in use at stations 1 and 2.

Assumption: interpret "difference" as

$$X = S_1 - S_2.$$

The smallest possible value is

$$0 - 4 = -4,$$

and the largest possible value is

$$6 - 0 = 6.$$

Therefore

$$X \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}.$$

Answer:

$$\boxed{\{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}}$$

(c) $U =$ the maximum number of pumps in use at either station.

$$U = \max(S_1, S_2)$$

The minimum possible maximum is 0, and the largest possible maximum is 6. Every integer from 0 to 6 can occur, so

$$U \in \{0, 1, 2, 3, 4, 5, 6\}.$$

Answer:

$$\boxed{\{0, 1, 2, 3, 4, 5, 6\}}$$

(d) $Z =$ the number of stations having exactly two pumps in use.

There are only two stations, so:

- $Z = 0$: neither station has exactly 2 pumps in use,
- $Z = 1$: exactly one station has 2 pumps in use,
- $Z = 2$: both stations have 2 pumps in use.

Thus

$$Z \in \{0, 1, 2\}.$$

Answer:

$$\boxed{\{0, 1, 2\}}$$

Problem 12

Let $Y =$ the number of ticketed passengers who show up for a flight with 50 seats. The pmf is:

y	45	46	47	48	49	50	51	52	53	54	55
$p(y)$	.05	.10	.12	.14	.25	.17	.06	.05	.03	.02	.01

(a) Probability that the flight will accommodate all ticketed passengers who show up.

All ticketed passengers can be accommodated when at most 50 passengers show up:

$$P(\text{all accommodated}) = P(Y \leq 50).$$

Thus,

$$P(Y \leq 50) = .05 + .10 + .12 + .14 + .25 + .17 = .83.$$

Answer:

$$\boxed{0.83}$$

(b) Probability that not all ticketed passengers who show up can be accommodated.

This occurs when more than 50 passengers show up:

$$P(\text{not all accommodated}) = P(Y > 50).$$

So,

$$P(Y > 50) = .06 + .05 + .03 + .02 + .01 = .17.$$

(Equivalently, $1 - .83 = .17$.)

Answer:

$$\boxed{0.17}$$

(c) Probability a standby passenger gets on the flight.

- If you are the *first* person on the standby list, you get on if at least 1 seat is available after ticketed passengers are seated. That requires

$$50 - Y \geq 1 \quad \Longleftrightarrow \quad Y \leq 49.$$

Therefore,

$$P(\text{first standby gets on}) = P(Y \leq 49)$$
$$= .05 + .10 + .12 + .14 + .25 = .66.$$

Answer:

$$\boxed{0.66}$$

- If you are the *third* person on the standby list, you get on if at least 3 seats are available. That requires

$$50 - Y \geq 3 \quad \Longleftrightarrow \quad Y \leq 47.$$

Hence,

$$P(\text{third standby gets on}) = P(Y \leq 47)$$
$$= .05 + .10 + .12 = .27.$$

Answer:

$$\boxed{0.27}$$

Problem 14

Let Y = the number of forms required, where $Y \in \{1, 2, 3, 4, 5\}$, and suppose

$$p(y) = ky, \qquad y = 1, 2, 3, 4, 5.$$

(a) Find k.

Since $p(y)$ is a pmf,

$$\sum_{y=1}^{5} p(y) = 1.$$

Thus,

$$\sum_{y=1}^{5} ky = k\sum_{y=1}^{5} y = 1.$$

Now

$$\sum_{y=1}^{5} y = 1+2+3+4+5 = 15,$$

so

$$15k = 1 \quad \Rightarrow \quad k = \frac{1}{15}.$$

Answer:

$$\boxed{k = \frac{1}{15}}$$

(b) Find $P(Y \leq 3)$.

$$P(Y \leq 3) = p(1) + p(2) + p(3)$$
$$= k(1) + k(2) + k(3) = k(1+2+3) = 6k.$$

Substituting $k = \frac{1}{15}$,

$$P(Y \leq 3) = 6\left(\frac{1}{15}\right) = \frac{6}{15} = \frac{2}{5} = 0.4.$$

Answer:

$$\boxed{P(Y \leq 3) = \frac{2}{5} = 0.4}$$

(c) Find $P(2 \leq Y \leq 4)$.

$$P(2 \leq Y \leq 4) = p(2) + p(3) + p(4)$$
$$= k(2+3+4) = 9k.$$

Using $k = \frac{1}{15}$,

$$P(2 \leq Y \leq 4) = 9\left(\frac{1}{15}\right) = \frac{9}{15} = \frac{3}{5} = 0.6.$$

Answer:

$$\boxed{P(2 \leq Y \leq 4) = \frac{3}{5} = 0.6}$$

(d) Could

$$p(y) = \frac{y^2}{50}, \qquad y = 1, 2, 3, 4, 5$$

be the pmf of Y?

For a valid pmf, the probabilities must sum to 1. Compute:

$$\sum_{y=1}^{5} \frac{y^2}{50} = \frac{1^2+2^2+3^2+4^2+5^2}{50} = \frac{1+4+9+16+25}{50} = \frac{55}{50} = 1.1.$$

Since $1.1 \neq 1$, this is not a valid pmf.

Answer:

$$\boxed{\text{No, because } \sum_{y=1}^{5} p(y) = \frac{55}{50} = 1.1 \neq 1.}$$

Problem 18

Two fair six-sided dice are tossed independently. Let

$$M = \max(\text{die } 1, \text{die } 2).$$

Since each die can show $1, \ldots, 6$, the possible values of M are

$$M \in \{1, 2, 3, 4, 5, 6\}.$$

There are 36 equally likely ordered outcomes.

(a) Find the pmf of M.

A convenient way is to count outcomes with maximum exactly m.

$$p_M(m) = P(M = m) = P(M \leq m) - P(M \leq m - 1).$$

Since both dice must be at most m,

$$P(M \leq m) = \left(\frac{m}{6}\right)^2 = \frac{m^2}{36}.$$

Therefore,

$$p_M(m) = \frac{m^2 - (m-1)^2}{36} = \frac{2m-1}{36}, \qquad m = 1, 2, 3, 4, 5, 6.$$

So the pmf is:

m	1	2	3	4	5	6
$p_M(m)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Equivalently,

$$p_M(1) = \frac{1}{36}, \quad p_M(2) = \frac{3}{36}, \quad p_M(3) = \frac{5}{36}, \quad p_M(4) = \frac{7}{36}, \quad p_M(5) = \frac{9}{36}, \quad p_M(6) = \frac{11}{36}.$$

Answer:

$$\boxed{p_M(m) = \frac{2m-1}{36}, \quad m = 1, 2, 3, 4, 5, 6}$$

(b) Determine the cdf of M and graph it.

The cdf is

$$F_M(m) = P(M \leq m).$$

Since both dice must be $\leq m$,

$$F_M(m) = \frac{m^2}{36}, \qquad m = 1, 2, 3, 4, 5, 6.$$

Writing it as a step function:

$$F_M(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{36}, & 1 \leq x < 2, \\ \frac{4}{36}, & 2 \leq x < 3, \\ \frac{9}{36}, & 3 \leq x < 4, \\ \frac{16}{36}, & 4 \leq x < 5, \\ \frac{25}{36}, & 5 \leq x < 6, \\ 1, & x \geq 6. \end{cases}$$

The cdf values at the integers are:

x	1	2	3	4	5	6
$F_M(x)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{9}{36}$	$\frac{16}{36}$	$\frac{25}{36}$	1

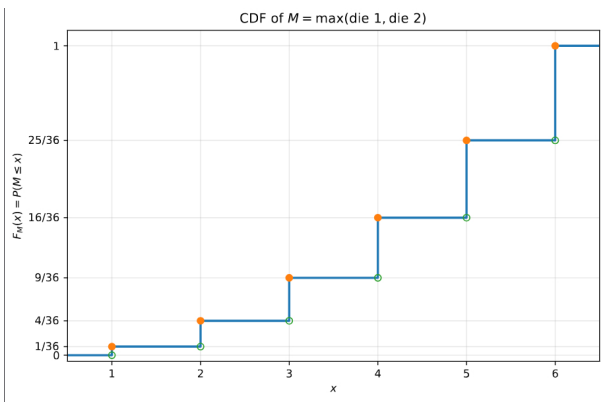

Figure 1: CDF graph for Problem 18(b)

Answer:

$$\boxed{F_M(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{36}, & 1 \leq x < 2, \\ \frac{4}{36}, & 2 \leq x < 3, \\ \frac{9}{36}, & 3 \leq x < 4, \\ \frac{16}{36}, & 4 \leq x < 5, \\ \frac{25}{36}, & 5 \leq x < 6, \\ 1, & x \geq 6. \end{cases}}$$

Problem 28

Let

$$F(x) = P(X \leq x)$$

be the cdf of a random variable X.

Suppose $x_1 < x_2$. Then the event

$$\{X \leq x_1\}$$

is contained in the event

$$\{X \leq x_2\},$$

because every value that is $\leq x_1$ is also $\leq x_2$. Thus

$$\{X \leq x_1\} \subseteq \{X \leq x_2\}.$$

Since probabilities preserve set inclusion,

$$P(X \leq x_1) \leq P(X \leq x_2).$$

Therefore,

$$F(x_1) \leq F(x_2),$$

so $F(x)$ is a nondecreasing function.

Also,

$$F(x_2) - F(x_1) = P(X \leq x_2) - P(X \leq x_1) = P(x_1 < X \leq x_2).$$

Hence,

$$F(x_1) = F(x_2) \quad \Longleftrightarrow \quad P(x_1 < X \leq x_2) = 0.$$

Answer:

$$\boxed{F(x) \text{ is nondecreasing because } x_1 < x_2 \implies \{X \leq x_1\} \subseteq \{X \leq x_2\}.}$$

Moreover,

$$\boxed{F(x_1) = F(x_2) \text{ exactly when } P(x_1 < X \leq x_2) = 0.}$$

Problem 32

The rated capacity X (in ft^3) has pmf

x	16	18	20
$p(x)$	0.2	0.5	0.3

(a) Compute $E(X)$, $E(X^2)$, and $V(X)$.

First,

$$E(X) = \sum x\, p(x) = 16(0.2) + 18(0.5) + 20(0.3).$$

$$E(X) = 3.2 + 9 + 6 = 18.2.$$

Next,

$$E(X^2) = \sum x^2 p(x) = 16^2(0.2) + 18^2(0.5) + 20^2(0.3).$$

$$E(X^2) = 256(0.2) + 324(0.5) + 400(0.3) = 51.2 + 162 + 120 = 333.2.$$

Then

$$V(X) = E(X^2) - [E(X)]^2 = 333.2 - (18.2)^2.$$

$$V(X) = 333.2 - 331.24 = 1.96.$$

Answer:

$$\boxed{E(X) = 18.2, \qquad E(X^2) = 333.2, \qquad V(X) = 1.96}$$

(b) If the price is $70X - 650$, find the expected price.

Let

$$Y = 70X - 650.$$

Using linearity of expectation,

$$E(Y) = E(70X - 650) = 70E(X) - 650.$$

Substituting $E(X) = 18.2$,

$$E(Y) = 70(18.2) - 650 = 1274 - 650 = 624.$$

Answer:

$$\boxed{624}$$

(c) Find the variance of the price paid.

Since $Y = 70X - 650$,

$$V(Y) = V(70X - 650) = 70^2 V(X).$$

Thus,

$$V(Y) = 4900(1.96) = 9604.$$

Answer:

$$\boxed{9604}$$

(d) If the actual capacity is

$$h(X) = X - 0.008X^2,$$

find the expected actual capacity.

We want

$$E[h(X)] = E(X - 0.008X^2) = E(X) - 0.008E(X^2).$$

Using the values from part (a),

$$E[h(X)] = 18.2 - 0.008(333.2).$$

$$E[h(X)] = 18.2 - 2.6656 = 15.5344.$$

Answer:

$$\boxed{15.5344 \text{ ft}^3}$$

Problem 36

Let X = damage incurred (in dollars), with pmf

x	0	1000	5000	10000
$p(x)$	0.8	0.1	0.08	0.02

The policy has a \$500 deductible, so the company's payment for a loss of size X is

$$Y = \begin{cases} 0, & X = 0, \\ 1000 - 500 = 500, & X = 1000, \\ 5000 - 500 = 4500, & X = 5000, \\ 10000 - 500 = 9500, & X = 10000. \end{cases}$$

Thus the expected payout is

$$E(Y) = 0(0.8) + 500(0.1) + 4500(0.08) + 9500(0.02).$$

Compute each term:

$$500(0.1) = 50, \qquad 4500(0.08) = 360, \qquad 9500(0.02) = 190.$$

So,

$$E(Y) = 50 + 360 + 190 = 600.$$

If the premium charged is P, then

$$\text{expected profit} = P - E(Y).$$

The company wants expected profit to be \$100, so

$$P - 600 = 100.$$

Therefore,

$$P = 700.$$

Answer:

$$\boxed{\$700}$$

Problem 42

Given

$$E(X) = 5 \qquad \text{and} \qquad E[X(X-1)] = 27.5.$$

(a) Find $E(X^2)$.

First verify the identity:

$$X(X-1) = X^2 - X.$$

Taking expectations,

$$E[X(X-1)] = E(X^2) - E(X).$$

Hence,

$$E(X^2) = E[X(X-1)] + E(X) = 27.5 + 5 = 32.5.$$

Answer:

$$\boxed{E(X^2) = 32.5}$$

(b) Find $V(X)$.

Use

$$V(X) = E(X^2) - [E(X)]^2.$$

Substituting the known values,

$$V(X) = 32.5 - 5^2 = 32.5 - 25 = 7.5.$$

Answer:

$$\boxed{V(X) = 7.5}$$

(c) Give the general relationship among $E(X)$, $E[X(X-1)]$, and $V(X)$.

Since

$$E[X(X-1)] = E(X^2) - E(X),$$

and

$$V(X) = E(X^2) - [E(X)]^2,$$

we can substitute $E(X^2) = E[X(X-1)] + E(X)$ into the variance formula:

$$V(X) = E[X(X-1)] + E(X) - [E(X)]^2.$$

Equivalently,

$$E[X(X-1)] = V(X) + [E(X)]^2 - E(X).$$

Answer:

$$\boxed{V(X) = E[X(X-1)] + E(X) - [E(X)]^2}$$

or equivalently

$$\boxed{E[X(X-1)] = V(X) + [E(X)]^2 - E(X)}.$$

Problem 48

Given

$$X \sim \text{Bin}(25, 0.05),$$

so

$$P(X = x) = \binom{25}{x}(0.05)^x(0.95)^{25-x}, \qquad x = 0, 1, \ldots, 25.$$

(a) Determine $P(X \leq 3)$ and $P(X < 3)$.

$$P(X \leq 3) = \sum_{x=0}^{3} \binom{25}{x} (0.05)^x (0.95)^{25-x}$$
$$= \binom{25}{0}(0.95)^{25} + \binom{25}{1}(0.05)(0.95)^{24} + \binom{25}{2}(0.05)^2(0.95)^{23} + \binom{25}{3}(0.05)^3(0.95)^{22}$$
$$\approx 0.9659.$$

Also,

$$P(X < 3) = P(X \leq 2)$$
$$= \sum_{x=0}^{2} \binom{25}{x} (0.05)^x (0.95)^{25-x} \approx 0.8729.$$

Answer:

$$\boxed{P(X \leq 3) \approx 0.9659, \qquad P(X < 3) \approx 0.8729}$$

(b) Determine $P(X \geq 4)$.

Use the complement:

$$P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.9659 = 0.0341.$$

Answer:

$$\boxed{P(X \geq 4) \approx 0.0341}$$

(c) Determine $P(1 \leq X \leq 3)$.

$$P(1 \leq X \leq 3) = P(X \leq 3) - P(X = 0).$$

Since

$$P(X = 0) = \binom{25}{0}(0.05)^0(0.95)^{25} = (0.95)^{25} \approx 0.2774,$$

we get

$$P(1 \leq X \leq 3) \approx 0.9659 - 0.2774 = 0.6885.$$

Answer:

$$\boxed{P(1 \leq X \leq 3) \approx 0.6885}$$

(d) What are $E(X)$ and σ_X?

For a binomial random variable,

$$E(X) = np, \qquad \sigma_X = \sqrt{np(1-p)}.$$

Thus

$$E(X) = 25(0.05) = 1.25,$$

and

$$\sigma_X = \sqrt{25(0.05)(0.95)} = \sqrt{1.1875} \approx 1.0897.$$

Answer:

$$\boxed{E(X) = 1.25, \qquad \sigma_X \approx 1.0897}$$

(e) In a sample of 50 children, what is the probability that none has a food allergy?

Here $X \sim \text{Bin}(50, 0.05)$, so

$$P(X = 0) = \binom{50}{0}(0.05)^0(0.95)^{50} = (0.95)^{50} \approx 0.0769.$$

Answer:

$$\boxed{0.0769}$$

Problem 50

Let X = the number of calls (out of 25) that involve a fax message. Then

$$X \sim \text{Bin}(25, 0.25),$$

so

$$P(X = x) = \binom{25}{x}(0.25)^x(0.75)^{25-x}, \qquad x = 0, 1, \ldots, 25.$$

(a) At most 6 of the calls involve a fax message.

$$P(X \leq 6) = \sum_{x=0}^{6} \binom{25}{x}(0.25)^x(0.75)^{25-x} \approx 0.5611.$$

Answer:

$$\boxed{P(X \leq 6) \approx 0.5611}$$

(b) Exactly 6 of the calls involve a fax message.

$$P(X = 6) = \binom{25}{6}(0.25)^6(0.75)^{19} \approx 0.1828.$$

Answer:

$$\boxed{P(X = 6) \approx 0.1828}$$

(c) At least 6 of the calls involve a fax message.

$$P(X \geq 6) = 1 - P(X \leq 5) \approx 0.6217.$$

Answer:

$$\boxed{P(X \geq 6) \approx 0.6217}$$

(d) More than 6 of the calls involve a fax message.

$$P(X > 6) = 1 - P(X \leq 6) = 1 - 0.5611 = 0.4389.$$

Answer:

$$\boxed{P(X > 6) \approx 0.4389}$$

Problem 52

Let X = the number of the 25 purchasers who want a new copy. Then

$$X \sim \text{Bin}(25, 0.30).$$

(a) Mean value and standard deviation.

For a binomial random variable,

$$\mu_X = E(X) = np, \qquad \sigma_X = \sqrt{np(1-p)}.$$

Thus

$$E(X) = 25(0.30) = 7.5,$$

and

$$\sigma_X = \sqrt{25(0.30)(0.70)} = \sqrt{5.25} \approx 2.2913.$$

Answer:

$$\boxed{E(X) = 7.5, \qquad \sigma_X \approx 2.2913}$$

(b) Probability that the number who want new copies is more than two standard deviations away from the mean.

We want

$$P\big(|X - \mu_X| > 2\sigma_X\big).$$

Since

$$\mu_X = 7.5, \qquad 2\sigma_X \approx 2(2.2913) = 4.5826,$$

this becomes

$$P(|X - 7.5| > 4.5826).$$

So

$$X < 7.5 - 4.5826 = 2.9174 \quad \text{or} \quad X > 7.5 + 4.5826 = 12.0826.$$

Because X is integer-valued, this is

$$P(X \leq 2 \text{ or } X \geq 13).$$

Therefore,

$$P(X \leq 2) + P(X \geq 13) = \sum_{x=0}^{2} \binom{25}{x} (0.3)^x (0.7)^{25-x} + \sum_{x=13}^{25} \binom{25}{x} (0.3)^x (0.7)^{25-x}.$$

Numerically,

$$P(|X - 7.5| > 2\sigma_X) \approx 0.0264.$$

Answer:

$$\boxed{0.0264}$$

(c) Probability that all 25 get the type of book they want if the store has 15 new and 15 used copies.

If X people want new copies, then $25 - X$ want used copies.

For everyone to get what they want, the store must have enough of each type:

$$X \leq 15 \qquad \text{and} \qquad 25 - X \leq 15.$$

The second inequality gives

$$X \geq 10.$$

So all 25 get what they want exactly when

$$10 \leq X \leq 15.$$

Hence

$$P(10 \leq X \leq 15) = \sum_{x=10}^{15} \binom{25}{x} (0.3)^x (0.7)^{25-x} \approx 0.1890.$$

Answer:

$$\boxed{0.1890}$$

(d) Expected total revenue from the sale of the next 25 copies.

If X of the 25 purchasers want new copies, then $25 - X$ want used copies. So the revenue is

$$h(X) = 100X + 70(25 - X).$$

Simplifying,

$$h(X) = 100X + 1750 - 70X = 30X + 1750.$$

Using the linearity of expectation,

$$E[h(X)] = E(30X + 1750) = 30E(X) + 1750.$$

Since $E(X) = 7.5$,

$$E[h(X)] = 30(7.5) + 1750 = 225 + 1750 = 1975.$$

Answer:

$$\boxed{\$1975}$$

Problem 54

Let X = the number of the next 10 customers who want the oversize version. Then

$$X \sim \text{Bin}(10, 0.60).$$

(a) Probability that at least six want the oversize version.

We want

$$P(X \geq 6) = \sum_{x=6}^{10} \binom{10}{x} (0.6)^x (0.4)^{10-x}.$$

Thus,

$$\begin{aligned} P(X \geq 6) = &\binom{10}{6}(0.6)^6(0.4)^4 + \binom{10}{7}(0.6)^7(0.4)^3 \\ &+ \binom{10}{8}(0.6)^8(0.4)^2 + \binom{10}{9}(0.6)^9(0.4) \\ &+ \binom{10}{10}(0.6)^{10}. \end{aligned}$$

Numerically,

$$P(X \geq 6) \approx 0.6331.$$

Answer:

$$\boxed{0.6331}$$

(b) Probability that the number who want the oversize version is within 1 standard deviation of the mean.

For a binomial random variable,

$$\mu_X = np = 10(0.6) = 6,$$

$$\sigma_X = \sqrt{np(1-p)} = \sqrt{10(0.6)(0.4)} = \sqrt{2.4} \approx 1.5492.$$

"Within 1 standard deviation of the mean" means

$$|X - 6| \leq 1.5492.$$

Since X is an integer, this gives

$$X \in \{5, 6, 7\}.$$

Therefore,

$$\begin{aligned} P(|X - \mu_X| \leq \sigma_X) &= P(5 \leq X \leq 7) \\ &= \sum_{x=5}^{7} \binom{10}{x}(0.6)^x(0.4)^{10-x}. \end{aligned}$$

Numerically,

$$P(5 \leq X \leq 7) \approx 0.6665.$$

Answer:

$$\boxed{0.6665}$$

(c) Probability that all of the next ten customers can get the version they want if the store has seven rackets of each version.

If X customers want oversize, then $10 - X$ want midsize.

For everyone to get what they want, both conditions must hold:

$$X \leq 7 \qquad \text{and} \qquad 10 - X \leq 7.$$

The second inequality gives

$$X \geq 3.$$

So all 10 customers can be satisfied exactly when

$$3 \leq X \leq 7.$$

Hence

$$P(3 \leq X \leq 7) = \sum_{x=3}^{7} \binom{10}{x} (0.6)^x (0.4)^{10-x}.$$

Numerically,

$$P(3 \leq X \leq 7) \approx 0.8204.$$

Answer:

$$\boxed{0.8204}$$

Problem 56

Let X = the number of the 25 selected students who received a special accommodation. Then

$$X \sim \text{Bin}(25, 0.02).$$

So

$$P(X = x) = \binom{25}{x} (0.02)^x (0.98)^{25-x}, \qquad x = 0, 1, \ldots, 25.$$

(a) Probability that exactly 1 student received a special accommodation.

$$P(X = 1) = \binom{25}{1} (0.02)^1 (0.98)^{24}.$$

$$P(X = 1) \approx 0.3079.$$

Answer:

$$\boxed{0.3079}$$

(b) Probability that at least 1 student received a special accommodation.
Use the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.98)^{25}.$$

$$P(X \geq 1) \approx 1 - 0.6035 = 0.3965.$$

Answer:

$$\boxed{0.3965}$$

(c) Probability that at least 2 students received a special accommodation.
Again use the complement:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

$$P(X \geq 2) = 1 - (0.98)^{25} - \binom{25}{1} (0.02)(0.98)^{24}.$$

$$P(X \geq 2) \approx 0.0886.$$

Answer:

$$\boxed{0.0886}$$

(d) Probability that the number accommodated is within 2 standard deviations of the expected number.

For a binomial random variable,

$$\mu_X = np = 25(0.02) = 0.5,$$

$$\sigma_X = \sqrt{np(1-p)} = \sqrt{25(0.02)(0.98)} = \sqrt{0.49} = 0.7.$$

Two standard deviations is

$$2\sigma_X = 1.4.$$

"Within 2 standard deviations of the mean" means

$$|X - 0.5| \leq 1.4.$$

So

$$-0.9 \leq X \leq 1.9.$$

Because X is integer-valued, this means

$$X = 0 \text{ or } X = 1.$$

Therefore,

$$P(|X - \mu_X| \leq 2\sigma_X) = P(X = 0) + P(X = 1).$$

$$P(X = 0) + P(X = 1) \approx 0.6035 + 0.3079 = 0.9114.$$

Answer:

$$\boxed{0.9114}$$

(e) Expected average time allowed for the 25 selected students.

If X students are accommodated, then $25 - X$ are not. Total time allowed is

$$T = 4.5X + 3(25 - X) = 4.5X + 75 - 3X = 75 + 1.5X.$$

Hence the average time per student is

$$\frac{T}{25} = \frac{75 + 1.5X}{25} = 3 + 0.06X.$$

Taking expectations,

$$E(3 + 0.06X) = 3 + 0.06E(X).$$

Since

$$E(X) = np = 25(0.02) = 0.5,$$

we get

$$E(\text{average time}) = 3 + 0.06(0.5) = 3.03.$$

Answer:

$$\boxed{3.03 \text{ hours}}$$

Problem 60

Let X = the number of passenger cars among the 25 vehicles. Then

$$X \sim \text{Bin}(25, 0.60),$$

so

$$E(X) = np = 25(0.60) = 15.$$

If X vehicles are passenger cars, then $25 - X$ are other vehicles. Thus the toll revenue is

$$h(X) = 1.00X + 2.50(25 - X).$$

Simplify:

$$h(X) = X + 62.5 - 2.5X = 62.5 - 1.5X.$$

Using linearity of expectation,

$$E[h(X)] = E(62.5 - 1.5X) = 62.5 - 1.5E(X).$$

Substitute $E(X) = 15$:

$$E[h(X)] = 62.5 - 1.5(15) = 62.5 - 22.5 = 40.$$

Answer:

$$\boxed{\$40.00}$$

Problem 66

Let Y = the number of reservation holders who actually appear for the trip.

Since each person with a reservation shows up with probability 0.8, when 6 reservations are made,

$$Y \sim \text{Bin}(6, 0.8).$$

(a) Probability that at least one person with a reservation cannot be accommodated.

The limousine holds at most 4 passengers, so at least one person cannot be accommodated when

$$Y \geq 5.$$

Therefore,

$$P(Y \geq 5) = P(Y = 5) + P(Y = 6).$$

Compute:

$$P(Y = 5) = \binom{6}{5}(0.8)^5(0.2) = 6(0.8)^5(0.2) = 0.393216,$$

$$P(Y = 6) = \binom{6}{6}(0.8)^6 = 0.262144.$$

Hence

$$P(Y \geq 5) = 0.393216 + 0.262144 = 0.65536.$$

Answer:

$$\boxed{0.65536}$$

(b) Expected number of available places when the limousine departs.

Let A = number of available places. Since the limousine has 4 seats,

$$A = \begin{cases} 4 - Y, & Y \leq 4, \\ 0, & Y \geq 5. \end{cases}$$

So

$$E(A) = \sum_{y=0}^{4} (4 - y) P(Y = y).$$

First compute:

$$P(Y = 0) = \binom{6}{0} (0.8)^0 (0.2)^6 = 0.000064,$$

$$P(Y = 1) = \binom{6}{1} (0.8)(0.2)^5 = 0.001536,$$

$$P(Y = 2) = \binom{6}{2} (0.8)^2 (0.2)^4 = 0.01536,$$

$$P(Y = 3) = \binom{6}{3} (0.8)^3 (0.2)^3 = 0.08192,$$

$$P(Y = 4) = \binom{6}{4} (0.8)^4 (0.2)^2 = 0.24576.$$

Then

$$E(A) = 4P(Y = 0) + 3P(Y = 1) + 2P(Y = 2) + P(Y = 3),$$

because the coefficient for $P(Y = 4)$ is 0. Thus

$$E(A) = 4(0.000064) + 3(0.001536) + 2(0.01536) + 1(0.08192) = 0.117504.$$

Answer:

$$\boxed{0.117504}$$

(c) Obtain the pmf of X, the number of passengers on a randomly selected trip.

Let N = number of reservations. The distribution of N is

$$P(N = 3) = 0.1, \quad P(N = 4) = 0.2, \quad P(N = 5) = 0.3, \quad P(N = 6) = 0.4.$$

Since the limousine can carry at most 4 passengers,

$$X = \min(S, 4),$$

where $S \mid N = n \sim \text{Bin}(n, 0.8)$.

Use the law of total probability:

$$P(X = x) = \sum_{n=3}^{6} P(X = x \mid N = n) P(N = n).$$

Conditional distributions:

If $N = 3$, then $X \sim \text{Bin}(3, 0.8)$:

$$P(X = 0 \mid N = 3) = 0.008, \qquad P(X = 1 \mid N = 3) = 0.096,$$
$$P(X = 2 \mid N = 3) = 0.384, \qquad P(X = 3 \mid N = 3) = 0.512.$$

If $N = 4$, then $X \sim \text{Bin}(4, 0.8)$:

$$P(X = 0 \mid N = 4) = 0.0016, \qquad P(X = 1 \mid N = 4) = 0.0256,$$
$$P(X = 2 \mid N = 4) = 0.1536, \qquad P(X = 3 \mid N = 4) = 0.4096,$$
$$P(X = 4 \mid N = 4) = 0.4096.$$

If $N = 5$, then

$$P(X = 0 \mid N = 5) = 0.00032, \qquad P(X = 1 \mid N = 5) = 0.0064,$$
$$P(X = 2 \mid N = 5) = 0.0512, \qquad P(X = 3 \mid N = 5) = 0.2048,$$
$$P(X = 4 \mid N = 5) = P(S \geq 4) = 0.73728.$$

If $N = 6$, then

$$P(X = 0 \mid N = 6) = 0.000064, \qquad P(X = 1 \mid N = 6) = 0.001536,$$
$$P(X = 2 \mid N = 6) = 0.01536, \qquad P(X = 3 \mid N = 6) = 0.08192,$$
$$P(X = 4 \mid N = 6) = P(S \geq 4) = 0.90112.$$

Therefore,

$$P(X = 0) = 0.1(0.008) + 0.2(0.0016) + 0.3(0.00032) + 0.4(0.000064) = 0.0012416,$$
$$P(X = 1) = 0.1(0.096) + 0.2(0.0256) + 0.3(0.0064) + 0.4(0.001536) = 0.0172544,$$
$$P(X = 2) = 0.1(0.384) + 0.2(0.1536) + 0.3(0.0512) + 0.4(0.01536) = 0.090624,$$
$$P(X = 3) = 0.1(0.512) + 0.2(0.4096) + 0.3(0.2048) + 0.4(0.08192) = 0.227328,$$
$$P(X = 4) = 0.2(0.4096) + 0.3(0.73728) + 0.4(0.90112) = 0.663552.$$

Thus the pmf of X is

x	0	1	2	3	4
$p(x)$	0.0012416	0.0172544	0.090624	0.227328	0.663552

Answer:

$$\boxed{p_X(x) = \begin{cases} 0.0012416, & x = 0, \\ 0.0172544, & x = 1, \\ 0.090624, & x = 2, \\ 0.227328, & x = 3, \\ 0.663552, & x = 4, \\ 0, & \text{otherwise.} \end{cases}}$$

Problem 68

There are $N = 18$ individuals total, of whom $K = 8$ are taking the test for the first time. A sample of $n = 6$ individuals is assigned to a particular examiner, and X is the number among these 6 who are first-time test takers.

(a) What distribution does X have?

Since X counts the number of "successes" in a sample of size 6 drawn *without replacement* from a population of size 18 containing 8 successes, X has a hypergeometric distribution:

$$X \sim \text{Hypergeometric}(N = 18,\ K = 8,\ n = 6).$$

Its pmf is

$$P(X = x) = \frac{\binom{8}{x}\binom{10}{6-x}}{\binom{18}{6}},$$

for

$$x = 0, 1, 2, 3, 4, 5, 6.$$

Answer:

$$\boxed{X \sim \text{Hypergeometric}(18, 8, 6)}$$

(b) Compute $P(X = 2)$, $P(X \leq 2)$, and $P(X \geq 2)$.

First,

$$P(X = 2) = \frac{\binom{8}{2}\binom{10}{4}}{\binom{18}{6}} = \frac{28 \cdot 210}{18564} = \frac{5880}{18564} = \frac{70}{221} \approx 0.3167.$$

Next,

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2).$$

So

$$P(X \leq 2) = \frac{\binom{8}{0}\binom{10}{6}}{\binom{18}{6}} + \frac{\binom{8}{1}\binom{10}{5}}{\binom{18}{6}} + \frac{\binom{8}{2}\binom{10}{4}}{\binom{18}{6}}.$$

$$P(X \leq 2) = \frac{210 + 2016 + 5880}{18564} = \frac{8106}{18564} \approx 0.4367.$$

Finally,

$$P(X \geq 2) = 1 - P(X \leq 1).$$

Thus

$$P(X \geq 2) = 1 - \left[\frac{\binom{8}{0}\binom{10}{6}}{\binom{18}{6}} + \frac{\binom{8}{1}\binom{10}{5}}{\binom{18}{6}}\right] \approx 0.8801.$$

Answer:

$$\boxed{P(X = 2) \approx 0.3167, \qquad P(X \leq 2) \approx 0.4367, \qquad P(X \geq 2) \approx 0.8801}$$

(c) Calculate the mean value and standard deviation of X.

For a hypergeometric random variable,

$$\mu_X = E(X) = n\frac{K}{N},$$

and

$$\sigma_X = \sqrt{n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}}.$$

Therefore,

$$E(X) = 6\left(\frac{8}{18}\right) = \frac{48}{18} = \frac{8}{3} \approx 2.6667.$$

Also,

$$\sigma_X = \sqrt{6\left(\frac{8}{18}\right)\left(\frac{10}{18}\right)\left(\frac{18-6}{18-1}\right)} = \sqrt{6\left(\frac{8}{18}\right)\left(\frac{10}{18}\right)\left(\frac{12}{17}\right)} \approx 1.0226.$$

Answer:

$$\boxed{E(X) = \frac{8}{3} \approx 2.6667, \qquad \sigma_X \approx 1.0226}$$

Problem 72

Let X = the number of the top four candidates who are interviewed on the first day.

There are $N = 11$ candidates total, with $K = 4$ "top" candidates, and $n = 6$ interview slots on the first day. Since the order is random, X has a hypergeometric distribution.

(a) Find the probability that x of the top four candidates are interviewed on the first day.

$$X \sim \text{Hypergeometric}(N = 11,\ K = 4,\ n = 6)$$

so

$$P(X = x) = \frac{\binom{4}{x}\binom{7}{6-x}}{\binom{11}{6}}, \qquad x = 0, 1, 2, 3, 4.$$

Thus the pmf is

$$P(X = 0) = \frac{\binom{4}{0}\binom{7}{6}}{\binom{11}{6}} = \frac{7}{462} \approx 0.0152,$$

$$P(X = 1) = \frac{\binom{4}{1}\binom{7}{5}}{\binom{11}{6}} = \frac{84}{462} \approx 0.1818,$$

$$P(X = 2) = \frac{\binom{4}{2}\binom{7}{4}}{\binom{11}{6}} = \frac{210}{462} \approx 0.4545,$$

$$P(X = 3) = \frac{\binom{4}{3}\binom{7}{3}}{\binom{11}{6}} = \frac{140}{462} \approx 0.3030,$$

$$P(X = 4) = \frac{\binom{4}{4}\binom{7}{2}}{\binom{11}{6}} = \frac{21}{462} \approx 0.0455.$$

Answer:

$$\boxed{P(X=x) = \frac{\binom{4}{x}\binom{7}{6-x}}{\binom{11}{6}}, \quad x = 0,1,2,3,4}$$

(b) How many of the top four candidates can be expected to be interviewed on the first day?

For a hypergeometric random variable,

$$E(X) = n\frac{K}{N}.$$

Therefore,

$$E(X) = 6\left(\frac{4}{11}\right) = \frac{24}{11} \approx 2.1818.$$

Answer:

$$\boxed{E(X) = \frac{24}{11} \approx 2.1818}$$

Problem 76

Let X = the number of children in the family. The family continues having children until it has three children of the same gender, where

$$P(B) = P(G) = 0.5.$$

The possible values of X are

$$X \in \{3,4,5\},$$

because:

- it is possible to stop after 3 children if the first three are all the same gender,
- it is possible to stop after 4 children if after three children the count is 2-to-1, and the fourth child matches the majority gender,
- if the family has not stopped by 4 children, then the first four must consist of 2 boys and 2 girls, so the fifth child guarantees three of one gender.

Now compute the probabilities.

$\boldsymbol{x = 3}$ This occurs if the first three children are all boys or all girls:

$$P(X=3) = P(BBB) + P(GGG) = \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 = \frac{2}{8} = \frac{1}{4}.$$

$\boldsymbol{x = 4}$ This occurs if among the first three children there are exactly 2 of one gender and 1 of the other, and the fourth child matches the majority gender.

There are 6 such patterns for the first three children:

$$BBG,\ BGB,\ GBB,\ GGB,\ GBG,\ BGG.$$

Each has probability $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, and then the fourth child must match the majority gender, which has probability $\frac{1}{2}$. Thus

$$P(X=4) = 6\left(\frac{1}{8}\right)\left(\frac{1}{2}\right) = \frac{6}{16} = \frac{3}{8}.$$

$\boldsymbol{x = 5}$ This occurs if the family has not stopped by 4 children. That requires exactly 2 boys and 2 girls among the first four children:

$$P(X = 5) = P(2 \text{ boys and } 2 \text{ girls in first } 4) = \binom{4}{2}\left(\frac{1}{2}\right)^4 = 6 \cdot \frac{1}{16} = \frac{3}{8}.$$

Therefore, the pmf of X is

$$p_X(x) = \begin{cases} \frac{1}{4}, & x = 3, \\ \frac{3}{8}, & x = 4, \\ \frac{3}{8}, & x = 5, \\ 0, & \text{otherwise.} \end{cases}$$

Check:

$$\frac{1}{4} + \frac{3}{8} + \frac{3}{8} = \frac{2}{8} + \frac{3}{8} + \frac{3}{8} = 1.$$

Answer:

$$\boxed{p_X(3) = \frac{1}{4}, \qquad p_X(4) = \frac{3}{8}, \qquad p_X(5) = \frac{3}{8}}$$

Problem 80

Let

$$X \sim \text{Poisson}(\mu = 4).$$

Then

$$P(X = x) = e^{-4}\frac{4^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

(a) Compute $P(X \leq 4)$ and $P(X < 4)$.

$$P(X \leq 4) = \sum_{x=0}^{4} e^{-4}\frac{4^x}{x!} = e^{-4}\left(1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!}\right) \approx 0.6288.$$

$$P(X < 4) = P(X \leq 3) = \sum_{x=0}^{3} e^{-4}\frac{4^x}{x!} \approx 0.4335.$$

Answer:

$$\boxed{P(X \leq 4) \approx 0.6288, \qquad P(X < 4) \approx 0.4335}$$

(b) Compute $P(4 \leq X \leq 8)$.

$$P(4 \leq X \leq 8) = \sum_{x=4}^{8} e^{-4}\frac{4^x}{x!} \approx 0.5452.$$

Answer:

$$\boxed{P(4 \leq X \leq 8) \approx 0.5452}$$

(c) Compute $P(8 \leq X)$.

$$P(X \geq 8) = 1 - P(X \leq 7) = 1 - \sum_{x=0}^{7} e^{-4} \frac{4^x}{x!} \approx 0.0511.$$

Answer:

$$\boxed{P(X \geq 8) \approx 0.0511}$$

(d) Probability that the number of anomalies exceeds its mean value by no more than one standard deviation.

For a Poisson random variable,

$$E(X) = \mu = 4, \qquad \sigma_X = \sqrt{\mu} = 2.$$

"Exceeds its mean value by no more than one standard deviation" means

$$4 < X \leq 4 + 2,$$

so, since X is integer-valued,

$$X = 5 \text{ or } 6.$$

Therefore,

$$P(5 \leq X \leq 6) = P(X = 5) + P(X = 6) = e^{-4} \frac{4^5}{5!} + e^{-4} \frac{4^6}{6!} \approx 0.2605.$$

Answer:

$$\boxed{0.2605}$$

Problem 82

Let

$$X \sim \text{Poisson}(\mu = 0.2).$$

Then the pmf is

$$P(X = x) = e^{-0.2} \frac{(0.2)^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

(a) Probability that a disk has exactly one missing pulse.

$$P(X = 1) = e^{-0.2} \frac{(0.2)^1}{1!} = 0.2e^{-0.2}.$$

Numerically,

$$P(X = 1) \approx 0.2(0.8187) = 0.1637.$$

Answer:

$$\boxed{P(X = 1) = 0.2e^{-0.2} \approx 0.1637}$$

(b) Probability that a disk has at least two missing pulses.

Use the complement:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

Now

$$P(X = 0) = e^{-0.2}, \qquad P(X = 1) = 0.2e^{-0.2}.$$

So

$$P(X \geq 2) = 1 - e^{-0.2} - 0.2e^{-0.2} = 1 - 1.2e^{-0.2}.$$

Numerically,

$$P(X \geq 2) \approx 1 - 1.2(0.8187) = 0.0175.$$

Answer:

$$\boxed{P(X \geq 2) = 1 - 1.2e^{-0.2} \approx 0.0175}$$

(c) If two disks are independently selected, find the probability that neither contains a missing pulse.

For one disk,

$$P(X = 0) = e^{-0.2}.$$

Because the two disks are independent,

$$P(\text{neither has a missing pulse}) = P(X = 0)^2 = (e^{-0.2})^2 = e^{-0.4}.$$

Numerically,

$$e^{-0.4} \approx 0.6703.$$

Answer:

$$\boxed{e^{-0.4} \approx 0.6703}$$

Problem 88

Let $X =$ the number of failed diodes on a board. Since each diode fails with probability $p = 0.01$ and there are $n = 200$ diodes,

$$X \sim \text{Bin}(200, 0.01).$$

Because n is large and p is small, we use the Poisson approximation with

$$\mu = np = 200(0.01) = 2,$$

so

$$X \approx \text{Poisson}(2).$$

(a) Expected number and standard deviation of the number that fail.

For a Poisson random variable,

$$E(X) = \mu, \qquad \sigma_X = \sqrt{\mu}.$$

Thus

$$E(X) = 2, \qquad \sigma_X = \sqrt{2} \approx 1.4142.$$

Answer:

$$\boxed{E(X) = 2, \qquad \sigma_X \approx 1.4142}$$

(b) Approximate probability that at least four diodes fail.

$$P(X \geq 4) = 1 - P(X \leq 3).$$

Using the Poisson pmf,

$$P(X \geq 4) = 1 - \sum_{x=0}^{3} e^{-2} \frac{2^x}{x!}.$$

So

$$P(X \geq 4) = 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!}\right) = 1 - e^{-2}\left(1 + 2 + 2 + \frac{4}{3}\right).$$

$$P(X \geq 4) \approx 0.1429.$$

Answer:

$$\boxed{P(X \geq 4) \approx 0.1429}$$

(c) If five boards are shipped, how likely is it that at least four work properly?

A board works properly only if no diodes fail, so for one board

$$P(\text{board works}) = P(X = 0) = e^{-2} \approx 0.1353.$$

Let Y = number of properly working boards among the 5. Then

$$Y \sim \text{Bin}(5, e^{-2}).$$

Therefore,

$$\begin{aligned} P(Y \geq 4) &= P(Y = 4) + P(Y = 5) \\ &= \binom{5}{4}(e^{-2})^4(1 - e^{-2}) + (e^{-2})^5. \end{aligned}$$

Numerically,

$$P(Y \geq 4) \approx 0.0015.$$

Answer:

$$\boxed{P(\text{at least 4 of 5 work properly}) \approx 0.0015}$$

Problem 2

Assume

$$X \sim \text{Uniform}(A = -5, B = 5).$$

Then the pdf is

$$f(x) = \begin{cases} \dfrac{1}{B - A} = \dfrac{1}{10}, & -5 \leq x \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

For a uniform distribution on $[a, b]$,

$$P(c < X < d) = \frac{d - c}{b - a}$$

for any interval $(c, d) \subseteq [a, b]$.

(a) Compute $P(X < 0)$.

The interval from -5 to 0 has length 5, and the total length is 10. Thus

$$P(X < 0) = \frac{0 - (-5)}{10} = \frac{5}{10} = \frac{1}{2}.$$

Answer:

$$\boxed{\frac{1}{2}}$$

(b) Compute $P(-2.5 < X < 2.5)$.

The interval length is

$$2.5 - (-2.5) = 5.$$

So

$$P(-2.5 < X < 2.5) = \frac{5}{10} = \frac{1}{2}.$$

Answer:

$$\boxed{\frac{1}{2}}$$

(c) Compute $P(-2 \leq X \leq 3)$.

For a continuous random variable, endpoints do not affect probability, so

$$P(-2 \leq X \leq 3) = P(-2 < X < 3).$$

The interval length is

$$3 - (-2) = 5.$$

Hence

$$P(-2 \leq X \leq 3) = \frac{5}{10} = \frac{1}{2}.$$

Answer:

$$\boxed{\frac{1}{2}}$$

(d) For k satisfying $-5 < k < k + 4 < 5$, compute $P(k < X < k + 4)$.

The interval $(k, k + 4)$ has length 4. Therefore

$$P(k < X < k + 4) = \frac{4}{10} = \frac{2}{5}.$$

Answer:

$$\boxed{\frac{2}{5}}$$

Problem 4

The proposed pdf is

$$f(x;\theta) = \begin{cases} \dfrac{x}{\theta^2} e^{-x^2/(2\theta^2)}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that $f(x;\theta)$ is a legitimate pdf.

To be a legitimate pdf, we need:

(a) $f(x;\theta) \geq 0$ for all x,

(b) $\displaystyle\int_{-\infty}^{\infty} f(x;\theta)\, dx = 1.$

Since $x > 0$, $\theta^2 > 0$, and $e^{-x^2/(2\theta^2)} > 0$, we have

$$f(x;\theta) \geq 0.$$

Also,

$$\int_{-\infty}^{\infty} f(x;\theta)\, dx = \int_0^{\infty} \frac{x}{\theta^2} e^{-x^2/(2\theta^2)}\, dx.$$

Let

$$u = -\frac{x^2}{2\theta^2}.$$

Then

$$du = -\frac{x}{\theta^2}\, dx, \qquad -du = \frac{x}{\theta^2}\, dx.$$

Therefore,

$$\int_0^{\infty} \frac{x}{\theta^2} e^{-x^2/(2\theta^2)}\, dx = -\int_{u=0}^{u=-\infty} e^u\, du = \int_{-\infty}^{0} e^u\, du = [e^u]_{-\infty}^{0} = 1.$$

Hence $f(x;\theta)$ is a legitimate pdf.

Answer:

$$\boxed{f(x;\theta) \text{ is a valid pdf since } f(x;\theta) \geq 0 \text{ and } \int_{-\infty}^{\infty} f(x;\theta)\, dx = 1.}$$

(b) Suppose $\theta = 100$. Find $P(X \leq 200)$, $P(X < 200)$, and $P(X \geq 200)$.

For $x > 0$,

$$P(X \leq x) = \int_0^x \frac{t}{\theta^2} e^{-t^2/(2\theta^2)}\, dt = 1 - e^{-x^2/(2\theta^2)}.$$

Thus, with $\theta = 100$,

$$P(X \leq 200) = 1 - e^{-200^2/(2\cdot 100^2)} = 1 - e^{-2}.$$

So

$$P(X \leq 200) \approx 1 - 0.1353 = 0.8647.$$

Since X is continuous,

$$P(X < 200) = P(X \leq 200) = 1 - e^{-2} \approx 0.8647.$$

Also,

$$P(X \geq 200) = 1 - P(X < 200) = e^{-2} \approx 0.1353.$$

Answer:

$$\boxed{P(X \leq 200) = 1 - e^{-2} \approx 0.8647}$$

$$\boxed{P(X < 200) = 1 - e^{-2} \approx 0.8647}$$

$$\boxed{P(X \geq 200) = e^{-2} \approx 0.1353}$$

(c) Find $P(100 \leq X \leq 200)$ for $\theta = 100$.

$$P(100 \leq X \leq 200) = P(X \leq 200) - P(X \leq 100).$$

Using

$$F(x) = 1 - e^{-x^2/(2\theta^2)},$$

with $\theta = 100$,

$$P(100 \leq X \leq 200) = (1 - e^{-2}) - (1 - e^{-1/2}) = e^{-1/2} - e^{-2}.$$

Numerically,

$$P(100 \leq X \leq 200) \approx 0.6065 - 0.1353 = 0.4712.$$

Answer:

$$\boxed{e^{-1/2} - e^{-2} \approx 0.4712}$$

(d) Give an expression for $P(X \leq x)$.

The cdf is

$$F(x) = P(X \leq x) = \begin{cases} 0, & x \leq 0, \\ \displaystyle\int_0^x \frac{t}{\theta^2} e^{-t^2/(2\theta^2)}\, dt, & x > 0. \end{cases}$$

Evaluating the integral gives

$$F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-x^2/(2\theta^2)}, & x > 0. \end{cases}$$

Answer:

$$\boxed{P(X \leq x) = F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-x^2/(2\theta^2)}, & x > 0. \end{cases}}$$

Problem 6

The pdf is

$$f(x) = \begin{cases} k\big[1-(x-3)^2\big], & 2 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of $f(x)$.

On $2 \le x \le 4$, the graph is a downward-opening parabola:

$$f(x) = k\big[1-(x-3)^2\big].$$

It has zeros at

$$x = 2, \qquad x = 4,$$

and its maximum occurs at

$$x = 3, \qquad f(3) = k.$$

Also,

$$f(x) = 0 \quad \text{for } x < 2 \text{ and } x > 4.$$

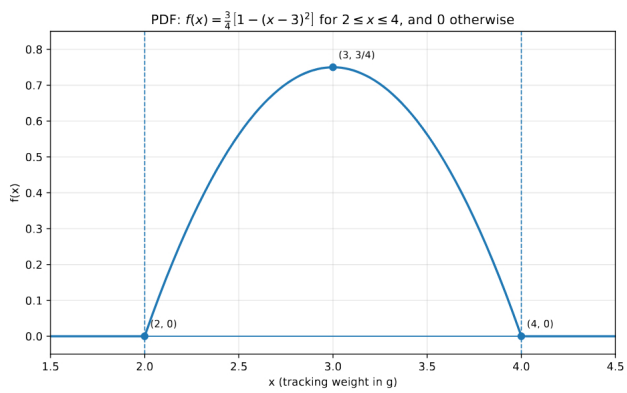

Figure 2: Sketch of $f(x)$ for Problem 6(a)

Answer: A downward-opening parabola on $[2,4]$, symmetric about $x = 3$, with endpoints $(2,0)$ and $(4,0)$, and peak at $(3,k)$; $f(x) = 0$ outside $[2,4]$.

(b) Find k.

Since $f(x)$ is a pdf,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Thus

$$\int_2^4 k\big[1-(x-3)^2\big]\,dx = 1.$$

Let

$$u = x - 3.$$

Then when $x = 2$, $u = -1$, and when $x = 4$, $u = 1$. So

$$k\int_{-1}^{1}(1-u^2)\,du = 1.$$

Now

$$\int_{-1}^{1}(1-u^2)\,du = \left[u - \frac{u^3}{3}\right]_{-1}^{1} = \left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right) = \frac{4}{3}.$$

Therefore,

$$k\left(\frac{4}{3}\right) = 1 \quad \Rightarrow \quad k = \frac{3}{4}.$$

Answer:

$$\boxed{k = \frac{3}{4}}$$

(c) What is the probability that the actual tracking weight is greater than the prescribed weight?

The prescribed weight is 3 g, so we want

$$P(X > 3).$$

Because the pdf is symmetric about $x = 3$,

$$P(X > 3) = P(X < 3) = \frac{1}{2}.$$

Answer:

$$\boxed{\frac{1}{2}}$$

(d) What is the probability that the actual weight is within 0.25 g of the prescribed weight?

We want

$$P(|X-3| \le 0.25) = P(2.75 \le X \le 3.25).$$

So

$$P(2.75 \le X \le 3.25) = \int_{2.75}^{3.25} \frac{3}{4}\left[1-(x-3)^2\right]\,dx.$$

Let $u = x - 3$. Then the limits become -0.25 to 0.25:

$$\frac{3}{4}\int_{-0.25}^{0.25}(1-u^2)\,du = \frac{3}{4}\left[u - \frac{u^3}{3}\right]_{-0.25}^{0.25}.$$

With $a = 0.25 = \frac{1}{4}$,

$$\left[u - \frac{u^3}{3}\right]_{-a}^{a} = 2\left(a - \frac{a^3}{3}\right) = 2\left(\frac{1}{4} - \frac{1}{64\cdot 3}\right) = 2\left(\frac{47}{192}\right) = \frac{47}{96}.$$

Hence

$$P(2.75 \le X \le 3.25) = \frac{3}{4}\cdot\frac{47}{96} = \frac{47}{128}.$$

Answer:

$$\boxed{\frac{47}{128} \approx 0.3672}$$

(e) What is the probability that the actual weight differs from the prescribed weight by more than 0.5 g?

We want

$$P(|X-3|>0.5) = 1 - P(|X-3| \le 0.5).$$

First compute

$$P(|X-3| \le 0.5) = P(2.5 \le X \le 3.5) = \int_{2.5}^{3.5} \frac{3}{4}\left[1-(x-3)^2\right] dx.$$

Using $u = x-3$, this becomes

$$\frac{3}{4}\int_{-0.5}^{0.5} (1-u^2)\, du = \frac{3}{4}\left[u - \frac{u^3}{3}\right]_{-0.5}^{0.5}.$$

With $a = 0.5 = \frac{1}{2}$,

$$\left[u - \frac{u^3}{3}\right]_{-a}^{a} = 2\left(a - \frac{a^3}{3}\right) = 2\left(\frac{1}{2} - \frac{1/8}{3}\right) = 2\left(\frac{1}{2} - \frac{1}{24}\right) = 2\left(\frac{11}{24}\right) = \frac{11}{12}.$$

Thus

$$P(|X-3| \le 0.5) = \frac{3}{4} \cdot \frac{11}{12} = \frac{11}{16}.$$

Therefore,

$$P(|X-3| > 0.5) = 1 - \frac{11}{16} = \frac{5}{16}.$$

Answer:

$$\boxed{\frac{5}{16} = 0.3125}$$

Problem 12

The cdf of X is

$$F(x) = \begin{cases} 0, & x < -2, \\ \frac{1}{2} + \frac{3}{32}\left(4x - \frac{x^3}{3}\right), & -2 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

(a) Compute $P(X < 0)$.

Since $0 \in [-2, 2)$,

$$P(X<0) = F(0) = \frac{1}{2} + \frac{3}{32}\left(0 - \frac{0^3}{3}\right) = \frac{1}{2}.$$

Answer:

$$\boxed{\frac{1}{2}}$$

(b) Compute $P(-1 < X < 1)$.

Because X is continuous,

$$P(-1 < X < 1) = F(1) - F(-1).$$

Now

$$F(1) = \frac{1}{2} + \frac{3}{32}\left(4 - \frac{1}{3}\right) = \frac{1}{2} + \frac{3}{32} \cdot \frac{11}{3} = \frac{1}{2} + \frac{11}{32} = \frac{27}{32},$$

and

$$F(-1) = \frac{1}{2} + \frac{3}{32}\left(-4 + \frac{1}{3}\right) = \frac{1}{2} - \frac{11}{32} = \frac{5}{32}.$$

Therefore,

$$P(-1 < X < 1) = \frac{27}{32} - \frac{5}{32} = \frac{22}{32} = \frac{11}{16}.$$

Answer:

$$\boxed{\frac{11}{16} = 0.6875}$$

(c) Compute $P(.5 < X)$.

Since X is continuous,

$$P(X > .5) = 1 - F(.5).$$

Compute

$$F(.5) = \frac{1}{2} + \frac{3}{32}\left(4 \cdot \frac{1}{2} - \frac{(.5)^3}{3}\right) = \frac{1}{2} + \frac{3}{32}\left(2 - \frac{1/8}{3}\right) = \frac{1}{2} + \frac{3}{32}\left(2 - \frac{1}{24}\right).$$

Thus

$$F(.5) = \frac{1}{2} + \frac{3}{32} \cdot \frac{47}{24} = \frac{1}{2} + \frac{47}{256} = \frac{175}{256}.$$

So

$$P(X > .5) = 1 - \frac{175}{256} = \frac{81}{256}.$$

Answer:

$$\boxed{\frac{81}{256} \approx 0.3164}$$

(d) Verify that $f(x)$ is as given in Exercise 3 by obtaining $F'(x)$.

Differentiate the middle piece:

$$F'(x) = \frac{3}{32}\left(4 - x^2\right), \qquad -2 < x < 2.$$

Also, outside $[-2, 2]$, $F(x)$ is constant, so

$$F'(x) = 0 \qquad \text{for } x < -2 \text{ and } x > 2.$$

Hence

$$f(x) = F'(x) = \begin{cases} \frac{3}{32}(4 - x^2), & -2 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Since

$$\frac{3}{32} = 0.09375,$$

this is exactly

$$f(x) = \begin{cases} 0.09375(4 - x^2), & -2 \le x \le 2, \\ 0, & \text{otherwise}, \end{cases}$$

which is the pdf from Exercise 3.

Answer:

$$\boxed{F'(x) = \frac{3}{32}(4 - x^2) \text{ on } (-2, 2), \text{ so this matches Exercise 3.}}$$

(e) Verify that $\tilde{\mu} = 0$.

Here $\tilde{\mu}$ denotes the median, so we must show

$$F(\tilde{\mu}) = \frac{1}{2}.$$

Since

$$F(0) = \frac{1}{2} + \frac{3}{32}\left(0 - \frac{0^3}{3}\right) = \frac{1}{2},$$

it follows that

$$\tilde{\mu} = 0.$$

Answer:

$$\boxed{\tilde{\mu} = 0}$$

Problem 14

Assume the depth X has a uniform distribution on $(7.5, 20)$. Thus

$$X \sim \text{Uniform}(a = 7.5, b = 20),$$

with pdf

$$f(x) = \begin{cases} \dfrac{1}{20 - 7.5} = \dfrac{1}{12.5}, & 7.5 < x < 20, \\ 0, & \text{otherwise}. \end{cases}$$

(a) Mean and variance of depth.

For a uniform distribution on (a, b),

$$E(X) = \frac{a + b}{2}, \qquad V(X) = \frac{(b - a)^2}{12}.$$

Therefore,

$$E(X) = \frac{7.5 + 20}{2} = 13.75,$$

and

$$V(X) = \frac{(20 - 7.5)^2}{12} = \frac{12.5^2}{12} = \frac{156.25}{12} = \frac{625}{48} \approx 13.0208.$$

Answer:

$$\boxed{E(X) = 13.75, \qquad V(X) = \frac{625}{48} \approx 13.0208}$$

(b) Cdf of depth.

For a uniform distribution on (a, b),

$$F(x) = P(X \le x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$

Hence

$$F(x) = \begin{cases} 0, & x \le 7.5, \\ \dfrac{x-7.5}{12.5}, & 7.5 < x < 20, \\ 1, & x \ge 20. \end{cases}$$

Answer:

$$\boxed{F(x) = \begin{cases} 0, & x \le 7.5, \\ \dfrac{x-7.5}{12.5}, & 7.5 < x < 20, \\ 1, & x \ge 20. \end{cases}}$$

(c) Probability that observed depth is at most 10; between 10 and 15.

First,

$$P(X \le 10) = \frac{10-7.5}{12.5} = \frac{2.5}{12.5} = 0.2.$$

Next,

$$P(10 \le X \le 15) = \frac{15-10}{12.5} = \frac{5}{12.5} = 0.4.$$

Answer:

$$\boxed{P(X \le 10) = 0.2, \qquad P(10 \le X \le 15) = 0.4}$$

(d) Probability that the observed depth is within 1 standard deviation of the mean value; within 2 standard deviations.

First compute the standard deviation:

$$\sigma = \sqrt{V(X)} = \sqrt{\frac{625}{48}} = \frac{25}{4\sqrt{3}} \approx 3.6084.$$

The mean is

$$\mu = 13.75.$$

Within 1 standard deviation:

$$P(\mu - \sigma \le X \le \mu + \sigma).$$

Since this interval lies inside $(7.5, 20)$, its length is 2σ, so

$$P(|X - \mu| \le \sigma) = \frac{2\sigma}{12.5} = \frac{2(25/(4\sqrt{3}))}{12.5} = \frac{1}{\sqrt{3}} \approx 0.5774.$$

Within 2 standard deviations:

$$\mu - 2\sigma \approx 13.75 - 7.2169 = 6.5331, \qquad \mu + 2\sigma \approx 13.75 + 7.2169 = 20.9669.$$

This interval contains the entire support $(7.5, 20)$, so

$$P(|X - \mu| \leq 2\sigma) = 1.$$

Answer:

$$\boxed{P(|X - \mu| \leq \sigma) = \frac{1}{\sqrt{3}} \approx 0.5774, \qquad P(|X - \mu| \leq 2\sigma) = 1}$$

Problem 18

Let

$$X \sim \text{Uniform}(-1, 1),$$

so its pdf is

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The limiter output is

$$Y = \begin{cases} X, & |X| \leq 0.5, \\ 0.5, & X > 0.5, \\ -0.5, & X < -0.5. \end{cases}$$

(a) Compute $P(Y = .5)$.

The value $Y = 0.5$ occurs whenever $X > 0.5$ (and also at $X = 0.5$, which has probability 0). Thus

$$P(Y = 0.5) = P(X > 0.5).$$

Since X is uniform on $[-1, 1]$,

$$P(X > 0.5) = \frac{1 - 0.5}{1 - (-1)} = \frac{0.5}{2} = \frac{1}{4}.$$

Answer:

$$\boxed{P(Y = 0.5) = \frac{1}{4}}$$

(b) Obtain the cdf of Y and graph it.

The cdf is

$$F_Y(y) = P(Y \leq y).$$

For $y < -0.5$,

$$F_Y(y) = 0.$$

For $-0.5 \leq y < 0.5$, the event $\{Y \leq y\}$ consists of:

- all $X < -0.5$, which contribute the point mass at -0.5,

- and all $-0.5 \le X \le y$, where $Y = X$.

So

$$F_Y(y) = P(X \le y) = \frac{y-(-1)}{2} = \frac{y+1}{2}.$$

For $y \ge 0.5$,

$$F_Y(y) = 1.$$

Therefore,

$$F_Y(y) = \begin{cases} 0, & y < -0.5, \\ \dfrac{y+1}{2}, & -0.5 \le y < 0.5, \\ 1, & y \ge 0.5. \end{cases}$$

In particular,

$$F_Y(-0.5) = \frac{-0.5+1}{2} = \frac{1}{4}, \qquad \lim_{y \to 0.5^-} F_Y(y) = \frac{0.5+1}{2} = \frac{3}{4}, \qquad F_Y(0.5) = 1.$$

The graph has a jump from 0 to $\frac{1}{4}$ at $y = -0.5$, a straight line $F_Y(y) = \frac{y+1}{2}$ on $[-0.5, 0.5)$, and a jump from $\frac{3}{4}$ to 1 at $y = 0.5$.

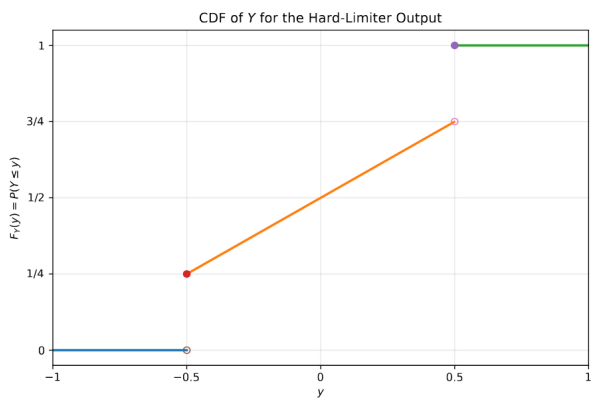

Figure 3: CDF graph for Problem 18(b)

Answer:

$$\boxed{F_Y(y) = \begin{cases} 0, & y < -0.5, \\ \dfrac{y+1}{2}, & -0.5 \le y < 0.5, \\ 1, & y \ge 0.5. \end{cases}}$$

Problem 22

The weekly demand X (in thousands of gallons) has pdf

$$f(x) = \begin{cases} 2\left(1 - \dfrac{1}{x^2}\right), & 1 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Compute the cdf of X.

For $x < 1$,

$$F(x) = 0.$$

For $1 \le x \le 2$,

$$F(x) = \int_1^x 2\left(1 - \frac{1}{t^2}\right)\,dt = 2\left[t + \frac{1}{t}\right]_1^x = 2\left(x + \frac{1}{x} - 2\right).$$

Thus

$$F(x) = 2x + \frac{2}{x} - 4, \qquad 1 \le x \le 2.$$

For $x > 2$,

$$F(x) = 1.$$

Therefore,

$$F(x) = \begin{cases} 0, & x < 1, \\ 2x + \dfrac{2}{x} - 4, & 1 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

Answer:

$$\boxed{F(x) = \begin{cases} 0, & x < 1, \\ 2x + \dfrac{2}{x} - 4, & 1 \le x \le 2, \\ 1, & x > 2. \end{cases}}$$

(b) Obtain an expression for the $(100p)$th percentile. What is the value of $\tilde{\mu}$?

Let x_p denote the $(100p)$th percentile, so

$$F(x_p) = p.$$

For $1 \le x_p \le 2$,

$$2x_p + \frac{2}{x_p} - 4 = p.$$

Multiply by x_p:

$$2x_p^2 - (p+4)x_p + 2 = 0.$$

Solving,

$$x_p = \frac{(p+4) \pm \sqrt{(p+4)^2 - 16}}{4} = \frac{(p+4) \pm \sqrt{p^2 + 8p}}{4}.$$

Since the percentile must lie in $[1, 2]$, we take the $+$ sign:

$$x_p = \frac{p + 4 + \sqrt{p^2 + 8p}}{4}.$$

The median $\tilde{\mu}$ is the 50th percentile, so set $p = 0.5$:

$$\tilde{\mu} = \frac{4.5 + \sqrt{4.25}}{4} = \frac{9 + \sqrt{17}}{8} \approx 1.6404.$$

Answer:

$$\boxed{x_p = \frac{p + 4 + \sqrt{p^2 + 8p}}{4}}$$

and

$$\boxed{\tilde{\mu} = \frac{9 + \sqrt{17}}{8} \approx 1.6404.}$$

(c) Compute $E(X)$ and $V(X)$.

First,

$$E(X) = \int_1^2 x\, 2\left(1 - \frac{1}{x^2}\right) dx = \int_1^2 2\left(x - \frac{1}{x}\right) dx.$$

So

$$E(X) = \left[x^2 - 2\ln x\right]_1^2 = (4 - 2\ln 2) - 1 = 3 - 2\ln 2.$$

Next,

$$E(X^2) = \int_1^2 x^2\, 2\left(1 - \frac{1}{x^2}\right) dx = \int_1^2 2(x^2 - 1)\, dx.$$

Hence

$$E(X^2) = 2\left[\frac{x^3}{3} - x\right]_1^2 = 2\left(\frac{8}{3} - 2 - \frac{1}{3} + 1\right) = \frac{8}{3}.$$

Therefore,

$$V(X) = E(X^2) - [E(X)]^2 = \frac{8}{3} - (3 - 2\ln 2)^2.$$

Answer:

$$\boxed{E(X) = 3 - 2\ln 2 \approx 1.6137}$$

$$\boxed{V(X) = \frac{8}{3} - (3 - 2\ln 2)^2 \approx 0.0620}$$

(d) If 1.5 thousand gallons are in stock at the beginning of the week and no new supply arrives, how much is expected to be left at the end of the week?

Let $h(x)$ = amount left when demand is x. Then

$$h(x) = \begin{cases} 1.5 - x, & 1 \le x \le 1.5, \\ 0, & 1.5 < x \le 2. \end{cases}$$

So

$$E[h(X)] = \int_1^{1.5} (1.5 - x)\, 2\left(1 - \frac{1}{x^2}\right) dx.$$

Expand the integrand:

$$(1.5 - x)\, 2\left(1 - \frac{1}{x^2}\right) = 3 - 2x - \frac{3}{x^2} + \frac{2}{x}.$$

Hence

$$E[h(X)] = \int_1^{3/2} \left(3 - 2x - \frac{3}{x^2} + \frac{2}{x}\right) dx$$

$$= \left[3x - x^2 + \frac{3}{x} + 2\ln x\right]_1^{3/2}.$$

Evaluating,

$$E[h(X)] = \left(\frac{9}{2} - \frac{9}{4} + 2 + 2\ln\frac{3}{2}\right) - 5 = -\frac{3}{4} + 2\ln\frac{3}{2}.$$

Answer:

$$\boxed{E[h(X)] = -\frac{3}{4} + 2\ln\left(\frac{3}{2}\right) \approx 0.0609}$$

thousand gallons.